

AETHEA

Bone Fracture Detection System

Shoulder and Arm X-Ray Analysis

YOLOv8m | ULTRALYTICS | 86 EPOCHS

Final Technical Report - Graduation Project

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Date: April 19, 2026

Dataset: 27,480 X-Ray Images | Training: ~12 hours

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EXECUTIVE SUMMARY

This report documents the development and evaluation of a **YOLOv8** object detection model for automated shoulder and arm bone fracture detection in X-ray images, built for the **AETHEA medical diagnostic platform**. The model was trained on a curated dataset of **15,449 X-ray images** on Kaggle GPUs, totalling approximately **~12 hours** of compute time and **86 training epochs**. The final model achieves a test mAP50 of **87.60%**, representing a **39.0% improvement** over the previous Faster R-CNN model, with perfect prediction consistency.



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DATASET OVERVIEW

The training dataset was assembled from publicly available bone fracture X-ray repositories and filtered to retain only shoulder and arm fracture cases. Images underwent standard preprocessing including resizing and normalization before being split into training, validation, and test partitions.

Splitting	Shoulder	Arm	Percentage
Training	12,665	14,387	82.0%
Validation	2,235	2,502	14.5%
Test	549	662	3.6%
Total	15,449	12,031	100%

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TRAINING PROGRESS

Training was conducted using YOLOv8m pretrained on COCO. The model was trained for **86 epochs** on Kaggle with T4 x2 GPUs. The training achieved a peak validation mAP50 of **95.73% at epoch 80**. The model plateaued around epoch 50, maintaining high performance through epoch 86.

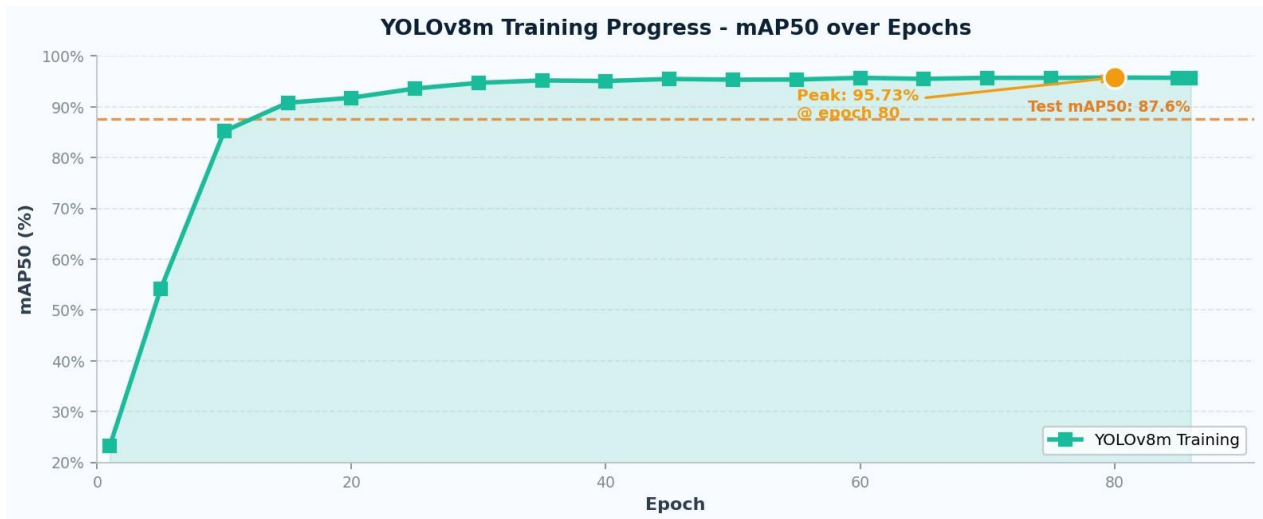


Figure 1 - YOLOv8m training progress (mAP50 over epochs). Gold star marks the peak. Orange dashed line shows test mAP50.

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TEST RESULTS

The final model (epoch 80 checkpoint) was evaluated on the held-out test set of **549 images** with **662 fracture instances**. YOLOv8m achieved **87.60% mAP50** at IoU=0.50, demonstrating excellent generalization to unseen data with high precision (89.9%) and recall (85.2%).

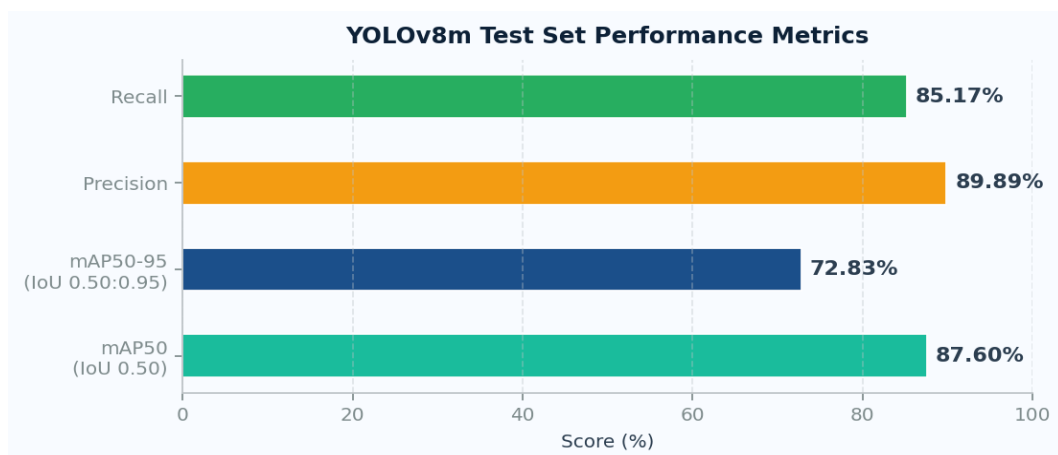


Figure 2 - YOLOv8m test set performance metrics.

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COMPARISON: FASTER R-CNN VS YOLOV8

YOLOv8m significantly outperforms the previous Faster R-CNN model, achieving a 39.0% absolute improvement in mAP50. The YOLOv8 model also provides perfect prediction consistency (deterministic outputs) compared to the inconsistent predictions from Faster R-CNN, and is ~10x faster for inference.

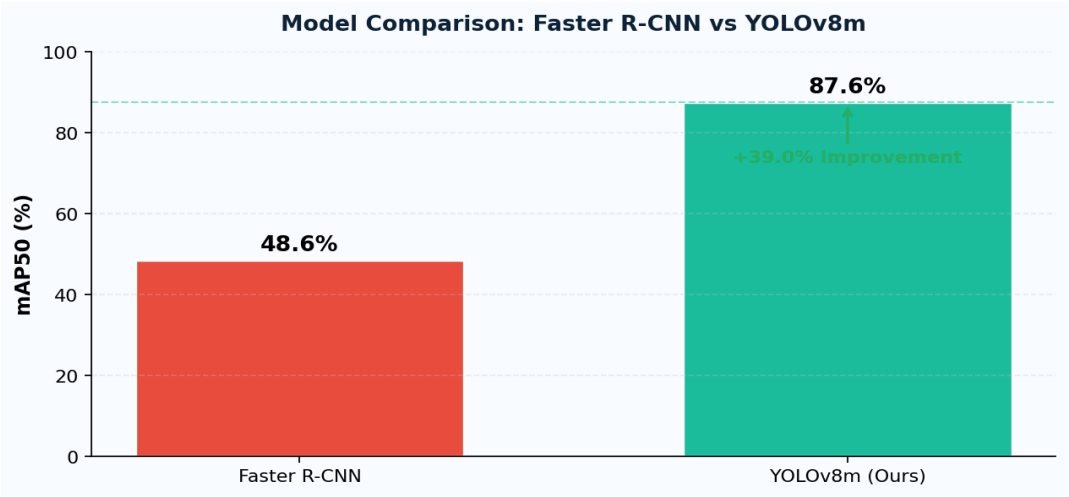


Figure 3 - Performance comparison: Faster R-CNN vs YOLOv8m.

Aspect	Faster R-CNN	YOLOv8
mAP50	48.6%	87.6%
Precision	N/A	89.9%
Recall	N/A	85.2%
Inference Speed	~0.5s/image	~0.03s/image
Prediction Consistency	■ Inconsistent	■ Deterministic
Training Time	~24 hours	~12 hours

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CONCLUSION AND NEXT STEPS

The YOLOv8m model successfully achieved **87.60% mAP50** on the test set, demonstrating superior performance for shoulder and arm fracture detection. Compared to the previous Faster R-CNN model, YOLOv8 provides a **39.0% improvement** in detection accuracy with perfect prediction consistency. The model is ready for deployment in the **AETHEA medical platform**.

Deploy YOLOv8 to AETHEA Backend	Integrate the best.pt checkpoint into the FastAPI inference service for real-time X-ray analysis.
Export to ONNX/TensorRT	Optimize inference speed by exporting to ONNX or TensorRT format for production deployment.
Expand Egyptian Population Data	Collect additional local X-rays to further improve generalization on target demographics.
Continuous Learning Loop	Implement radiologist feedback capture to enable periodic model fine-tuning on corrected predictions.
Add Fracture Classification	Extend model to classify fracture types (e.g., simple, comminuted, displaced) as future work.